

Ostbayerische Technische Hochschule Amberg/Weiden
Department of Electrical Engineering, Media, and
Computer Science

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Project Mobile & Ubiquitous Computing Summer
Semester 2026

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Courses of study: Artificial Intelligence, Industry-4.0-Informatics

June 9, 2026

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Abstract

1 Introduction

1.1 Intro

1.2 Mission Statement

1.3 Motivation

1.4 Document Structure

2 Related Work

3 Project Management

3.1 SMART

3.2 User Stories (Requirements)

3.3 MVP

3.4 Extensions

3.5 Task Distribution

4 Technical Concept

4.1 High-Level Architecture

4.2 Components

5 Hardware

Written in Arduino. [?]

5.1 Components (ESP32) in Detail

ESP32: [?]

5.2 Sensor Data Acquisition

5.3 Physics Simulation

5.4 Display

5.5 MQTT Topics & Payloads

6 Node-RED

[?]

6.1 Workflow

6.2 Dashboard & User Input

7 Android App

7.1 Software Components, Classes, Views

7.2 Dashboard & User Input

8 Evaluation

8.1 Problems & Solutions

[?] should not be used anymore. So we decided on using [?] instead for our dashboard.

8.2 Code Coverage

8.3 Cost Estimation

9 Summary

9.1 Summary & Future Work

A Prompts

B ESP32 Code